

1. View direction

$$\mathbf{d} = \frac{\mu - \mathbf{o}_{cam}}{\|\mu - \mathbf{o}_{cam}\|}$$
$$(x, y, z) = \mathbf{d}$$

2. Basis functions

Real SH polynomials in x, y, z up to degree 3, e.g. xy and $3xx - yy$

3. Accumulate by degree

$\mathbf{c} = C_0 \text{sh}_{dc}$, then add bands $l = 1, 2, 3$ with their C_l

4. Offset and clamp

$$\mathbf{c}_{final} = \max(\mathbf{c} + 0.5, 0)$$

Runs per Gaussian. Flags $\text{clamped}[3i + k] = (c_k < 0)$ are stored so backward applies zero gradient on clamped channels.